United States Patent Application Publication	20250285545
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	SEKI; Tatenobu et al.

INFORMATION PROVIDING APPARATUS, INFORMATION PROVIDING METHOD, AND NON-TRANSITORY COMPUTER-READABLE RECORDING MEDIUM

Abstract

A server device acquires operation data for identifying operation that is performed on a controller by a worker in accordance with a work instruction on a predetermined work process related to the controller that is operated by the worker, analyzes the operation data, determines a notification content for the worker, and notifies the worker of the determined notification content.

Inventors: **SEKI; Tatenobu (Tokyo, JP), HIROTANI; Kazutaka (Tokyo, JP), EMA; Nobuaki (Tokyo, JP)**

Applicant: **Yokogawa Electric Corporation** (Tokyo, JP)

Family ID: **1000008494343**

Appl. No.: **19/067317**

Filed: **February 28, 2025**

Foreign Application Priority Data

JP	2024-036332	Mar. 08, 2024

Publication Classification

Int. Cl.: **G09B5/02** (20060101); **G06Q10/0631** (20230101); **G06V20/52** (20220101); **G06V30/14** (20220101); **G06V40/20** (20220101)

U.S. Cl.:

CPC **G09B5/02** (20130101); **G06Q10/063114** (20130101); **G06V20/52** (20220101); **G06V30/14** (20220101); **G06V40/20** (20220101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to and incorporates by reference the entire contents of Japanese Patent Application No. 2024-036332 filed in Japan on Mar. 8, 2024.

FIELD

[0002] The present invention relates to an information providing apparatus, an information providing method, and a non-transitory computer-readable recording medium.

BACKGROUND

[0003] In a production plant, most of controllers in each of which a Programmable Logic Controller (PLC) or the like is incorporated are operated by installed operation panels of touch monitor types, and are independent without being connected to a network. A worker who operates the controllers that are not connected to the network need to operate the operation panels of the touch monitor types in the respective controllers (see, for example, Japanese Unexamined Patent Application Publication No. 2002-041124 and Japanese Unexamined Patent Application Publication No. 2014-071683).

[0004] However, in the conventional technology, it is difficult to connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network. For example, in the conventional technology, a worker who operates a controller that is not connected to a network in a production plant need to check, in advance, Standard Operation Procedure (SOP) of each of products, and perform manual operation.

[0005] The present invention has been conceived in view of the foregoing situations, and an object of the present invention is to connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

SUMMARY

[0006] According to an aspect of the embodiments, an information providing apparatus includes an acquisition unit that acquires operation data for identifying operation that is performed on an operation device by a worker in accordance with a work instruction on a predetermined work process related to the operation device that is operated by the worker, an analysis unit that analyzes the operation data and determines a notification content for the worker, and a notification unit that notifies the worker of the determined notification content.

[0007] According to an aspect of the embodiments, an information providing method that is implemented by a computer, the information providing method includes acquiring operation data for identifying operation that is performed on an operation device by a worker in accordance with a work instruction on a predetermined work process related to the operation device that is operated by the worker, analyzing the operation data, determining a notification content for the worker, and notifying the worker of the determined notification content.

[0008] According to an aspect of the embodiments, a non-transitory computer-readable recording medium having stored therein an information providing that causes a computer to execute a process, the process includes acquiring operation data for identifying operation that is performed on an operation device by a worker in accordance with a work instruction on a predetermined work process related to the operation device that is operated by the worker, analyzing the operation data, determining a notification content for the worker, and notifying the worker of the determined notification content.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a diagram illustrating a configuration example and a process example of a work management system according to one embodiment;

[0010] FIG. **2** is a block diagram illustrating a configuration example of each of apparatuses in the work management system according to one embodiment;

[0011] FIG. **3** is a diagram illustrating an example of an instruction data storage unit of a server device according to one embodiment;

[0012] FIG. **4** is a diagram illustrating an example of an operation data storage unit of the server device according to one embodiment;

[0013] FIG. **5** is a diagram illustrating an example of an analysis result storage unit of the server device according to one embodiment;

[0014] FIG. **6** is a diagram illustrating an example of a work report storage unit of the server device according to one embodiment;

[0015] FIG. **7** is a diagram illustrating a specific example of a display screen of a worker terminal according to one embodiment;

[0016] FIG. **8** is a diagram illustrating a specific example of a display screen of a controller according to one embodiment;

[0017] FIG. **9** is a flowchart illustrating an example of the entire flow of the work management system according to one embodiment;

[0018] FIG. **10** is a flowchart illustrating an example of the flow of a work instruction notification process of the work management system according to one embodiment;

[0019] FIG. **11** is a flowchart illustrating an example of the flow of an operation data acquisition process of the work management system according to one embodiment;

[0020] FIG. **12** is a flowchart illustrating an example of an operation data analysis process of the work management system according to one embodiment;

[0021] FIG. **13** is a flowchart illustrating an example of the flow of a work report generation process of the work management system according to one embodiment;

[0022] FIG. **14** is a diagram illustrating a configuration example and a process example of a work management system according to a first modification of one embodiment;

[0023] FIG. **15** is a diagram illustrating a specific example of a display screen of smart glasses according to the first modification of one embodiment;

[0024] FIG. **16** is a diagram illustrating a configuration example and a process example of a work management system according to a second modification of one embodiment;

[0025] FIG. **17** is a diagram illustrating a configuration example and a process example of a work management system according to a third modification of one embodiment;

[0026] FIG. **18** is a diagram illustrating a configuration example of a work management system according to a fourth modification of one embodiment; and

[0027] FIG. **19** is a diagram for explaining a hardware configuration example according to one embodiment;

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0028] Embodiments of an information providing apparatus, an information providing method, and a non-transitory computer-readable recording medium according to the present invention will be described in detail below with reference to the drawings. The present invention is not limited by the embodiments described below.

[0029] A configuration and a process of a work management system **100** according to one embodiment, a configuration and a process of each of apparatuses in the work management system **100**, flows of processes of the work management system **100**, modifications and application examples of one embodiment, and effects of one embodiment will be described below. Meanwhile, in one embodiment, a work site of a production plant will be described as one example, but a field of use is not specifically limited. For example, embodiments are applicable to an environment in which an operation apparatus that is an apparatus to be operated by a worker W and that is not connected to a network is present.

1. CONFIGURATION AND PROCESS OF WORK MANAGEMENT SYSTEM **100**

[0030] A configuration and a process of the work management system **100** according to one embodiment will be described in detail below with reference to FIG. **1**. FIG. **1** is a diagram illustrating a configuration example and a process example of the work management system **100** according to one embodiment. In the following, an entire configuration example of the work management system **100**, a process example of the work management system **100**, and an effect of the work management system **100** will be described.

1-1. Entire Configuration Example of Work Management System **100**

[0031] The work management system **100** includes a server device **10**, a worker terminal **20**, and a controller **30**. Here, the server device **10** and the worker terminal **20** are communicably connected to each other in a wired or wireless manner via a predetermined communication network (not illustrated). Meanwhile, it is possible to adopt, as the predetermined communication network, various kinds of communication networks, such as the Internet or a dedicated line.

1-1-1. Server Device **10**

[0032] The server device **10** is an information providing apparatus that gives a work instruction to a worker W and generates a work report. For example, the server device **10** is implemented by a cloud environment, an on-premise environment, an edge environment, or the like. Meanwhile, the work management system **100** illustrated in FIG. **1** may include the plurality of server devices **10**.

1-1-2. Worker Terminal **20**

[0033] The worker terminal **20** is an example of an apparatus that is operated by the worker W and that is connected to the network. In the example illustrated in FIG. **1**, the worker terminal **20** is a terminal that is carried by the worker W in a work site of a production plant, is implemented by a tablet terminal, a smartphone, or the like, and includes a monitor M that displays the work instruction. Meanwhile, the work management system **100** illustrated in FIG. **1** may include the plurality of worker terminals **20**.

1-1-3. Controller **30**

[0034] The controller **30** is an example of a device (operation device) that is operated by the worker W and that is not connected to the network. In the example illustrated in FIG. **1**, the controller **30** is an apparatus that is operated by the worker W in a work site of a production plant and is an apparatus that controls a control target apparatus (for example, a tank, a pump, or a conveyor) in the production plant. The controller **30** includes an operation panel P that receives operation performed by the worker W.

1-1-4. Camera C

[0035] The camera C is an image capturing device that captures an image of the operation panel P of the controller **30**, and is installed at a location at which the image of the operation panel P can be captured around the controller **30**. Here, the server device **10** and the camera C are communicably connected to each other in a wired or wireless manner via a predetermined communication network (not illustrated).

1-1-5. Eye-Tracking Sensor S

[0036] An eye-tracking sensor S is a sensor device that detects a line of sight of the worker W, and is installed at a location at which the line of sight of the worker W is detectable around the controller **30**. Here, the server device **10** and the eye-tracking sensor S are communicably connected to each other in a wired or wireless manner via a predetermined communication network (not illustrated).

1-2. Example of Overall Process of Work Management System **100**

[0037] An overall process performed by the work management system **100** as described above will be described. Meanwhile, processes from Step S**1** to Step S**5** below may be performed in a different order. Further, any of the processes from Step S**1** to Step S**5** below may be omitted.

1-2-1. Device Operation Process

[0038] Firstly, the worker W operates the controller **30** (Step S**1**). For example, the worker W operates the operation panel P based on the SOP. Further, the worker W performs a manual work, such as raw material input, and a work, such as data recording or data check during automatic control on the controller **30**.

1-2-2. Operation Data Acquisition Process

[0039] Secondly, the server device **10** acquires operation data (Step S**2**). For example, the server device **10** acquires, as the operation data, image data of the operation panel P that is captured by the camera C, line-of-sight data of the worker W that is detected by the eye-tracking sensor S, location data of the worker W that is detected by the worker terminal **20**, or the like.

1-2-3. Operation Data Analysis Process

[0040] Thirdly, the server device **10** analyzes the operation data (Step S**3**). For example, when a work process is operation of the operation panel P, the server device **10** analyzes the image data of the operation panel P and determines whether or not the worker W has operated the operation panel P correctly. In this case, the server device **10** determines appropriateness of the operation of the worker W based on layout information on the operation screen of the operation panel P or information on reply, transition, or the like at the time of pressing a button of the operation panel P.

[0041] Further, when the work process is the data recording, the server device **10** analyzes the eye-tracking data of the worker W and records characters in the operation panel P that is viewed by the worker W. In this case, the server device **10** identifies a region of the operation panel P that is in the line of sight of the worker W, performs an Optical Character Recognition (OCR) process on image data of the operation panel P that is included in the identified region, and identifies the characters in the operation panel P that is viewed by the worker W. Furthermore, it may be possible to directly perform the OCR process on the image data of the operation panel P and identify the characters in the operation panel P without using the eye-tracking data of the worker W that is detected by the eye-tracking sensor S.

[0042] Moreover, the server device **10** is able to analyze the location data of the worker W and determine whether or not the worker W operates the correct controller **30**.

[0043] Furthermore, the server device **10** determines a notification content based on an analysis result of the operation data. For example, when determining that the worker W has performed correct operation, the server device **10** determines a second work instruction as the notification content. In contrast, when determining that the worker W has performed incorrect operation, the server device **10** determines an error message as the notification content. Moreover, when determining that the worker W has performed incorrect operation, the server device **10** may determine a changed work instruction as the notification content.

1-2-4. Work Instruction Notification Process

[0044] Fourthly, the server device **10** notifies the worker W of the second work instruction (Step S**4**). For example, when determining that the worker W has performed correct operation at Step S**3** as described above, the server device **10** transmits a next work instruction among work processes that are generated based on the SOP to the worker terminal **20**, and displays the work instruction on the monitor M of the worker terminal **20**.

[0045] In this case, the server device **10** notifies the worker W of, as the work instruction, an instruction on operation of the operation panel P that is operation on the controller **30**, an instruction on a manual work, such as raw material input, an instruction on data recording, or the like. Further, the worker terminal **20** may display a flowchart that indicates a work instruction of each of work processes, and display a work instruction that is being carried out in a highlighted manner. Furthermore, the worker terminal **20** may display a list of the work instructions of all of the work processes, and display a work instruction that is being carried out in a highlighted manner.

[0046] In contrast, when determining that the worker W has performed incorrect operation at Step

S**3** as described above, the server device **10** gives a notice of the error message, the changed work instruction, or the like.

[0047] The processes from Step S**1** to Step S**4** as described above are repeated until all of the work processes that are performed by the controller **30** are completed. Meanwhile, in the processes from Step S**1** to Step S**4** as described above, the worker W is notified of the work instruction on the next work process after an operation data process at Step S**3**, but it may be possible to notify the worker W of a work instruction of a work process to be next performed before a device operation process at Step S**1**. Furthermore, in a work instruction notification process at Step S**4** as described above, it may be possible not to give a notice of the work instruction when determining that the worker W has performed correct operation, and it may be possible to give a notice of the error message, the changed work instruction, or the like when determining that the worker W has performed incorrect operation.

1-2-5. Work Report Generation Process

[0048] Fifthly, the server device **10** generates a work report (Step S**5**). For example, when all of the work processes that are performed by the controller **30** are completed, the server device **10** generates a work report that includes an identification number of the controller **30**, a name of the worker W, a work content, recorded data, or the like.

1-3. Effect of Work Management System **100**

[0049] In the following, an overview and a problem with a work management system **100**P according to a reference technology will be described, and thereafter, an effect of the work management system **100** will be described.

1-3-1. Overview of Work Management System **100**P

[0050] An overview of the work management system **100**P according to the reference technology will be described. In the work management system **100**P, in a production plant, the controller **30** in which a PLC or the like is incorporated is operated by the installed operation panel P of a touch monitor type, and is independent without being connected to a network. Therefore, in the work management system **100**P, in the production plant that includes the controller **30** that is not connected to the network, the worker W who operates the controller **30** needs to operate the operation panels P in the respective controllers **30**.

1-3-2. Problems with Work Management System **100***p*

[0051] Problems with the work management system **100**P the reference technology will be described. The work management system **100**P has first to fourth problems as described below.

1-3-2-1. First Problem

[0052] In the work management system **100**P, for example, as for a batch process in a production plant, because it is needed to manufacture various types of products, manufacturing procedures for the respective products are different and are not routine works. Therefore, in the work management system **100**P, the worker W needs to check the SOP for each of the products in advance and then perform manual operation.

1-3-2-2. Second Problem

[0053] In the work management system **100**P, work record, record of measurement values, and the like are performed by manual work by writing by hand on paper. Therefore, in the work management system **100**P, data is not digitalized and data utilization has not progressed.

1-3-2-3. Third Problem

[0054] In the work management system **100**P, to collect digital data, a system modification is needed to connect each of the controllers **30** to a network. However, to perform the modification as described above on existing facilities in the work management system **100**P, it is needed to obtain PLC information from a PLC vendor who has performed engineering on the PLC and the controllers **30**, and even products of the same PLC vendor may have different specifications depending on series. Accordingly, the above-described modification of the work management system **100**P needs a large amount of period and cost and imposes a burden on the work site, and therefore, it is difficult for a small-scale apparatus manufacturer, a customer, or the like to perform the modifications.

1-3-2-4. Fourth Problem

[0055] In the work management system **100**P, some of the controllers **30** may include, inside thereof, black boxes including know-how or the like of the apparatus manufacturer. Therefore, a system modification for connecting each of the controllers **30** in the work management system **100**P to a network has a problem in terms of confidentiality because the system modification includes installation of an electronic interface with an external apparatus.

1-3-3. Overview of Work Management System **100**

[0056] An overview of the work management system **100** according to one embodiment will be described. The work management system **100** performs processes as described below. In the following, overviews of a device operation process, an operation data acquisition process, an operation data analysis process, a work instruction notification process, and a work report generation process will be described.

1-3-3-1. Device Operation Process

[0057] Firstly, the worker W operates the controller **30**. In this case, the worker W performs a certain work, such as operation on the operation panel P, a manual work, data recording, or a check of data of the controller **30**, for example.

1-3-3-2. Operation Data Acquisition Process

[0058] Secondly, the server device **10** acquires operation data. In this case, the server device **10** acquires, as the operation data, image data of the operation panel P that is captured by the camera C, eye-tracking data of the worker W that is detected by the eye-tracking sensor S, or location data of the worker W that is detected by the worker terminal **20**, for example.

1-3-3-3. Operation Data Analysis Process

[0059] Thirdly, the server device **10** analyzes the operation data. In this case, the server device **10** analyzes the image data of the operation panel P and determines whether or not the worker W has correctly operated the operation panel P, for example. Further, the server device **10** analyzes the eye-tracking data of the worker W and records data of the operation panel P that is in the line of sight of the worker W, for example. Furthermore, the server device **10** may also analyze the location data of the worker W and determines whether or not the worker W is operating the correct controller **30**, for example. Then, the server device **10** determines a notification content based on an analysis result of the operation data.

1-3-3-4. Work Instruction Notification Process

[0060] Fourthly, the server device **10** notifies the worker W of the work instruction. In this case, when determining that the worker W has performed correct operation, the server device **10** transmits a next work instruction among the work processes that are generated based on the SOP to the worker terminal **20**, and displays the work instruction on the monitor M of the worker terminal **20**, for example. In contrast, when determining that the worker W has performed incorrect operation, the server device **10** gives a notice of the error message, the changed work instruction, or the like, for example.

1-3-3-5. Work Report Generation Process

[0061] Fifthly, the server device **10** generates a work report. In this case, when all of the work processes that are performed by the controller **30** are completed, the server device **10** generates a work report that indicates a work status, for example.

1-3-4. Effects of the Work Management System **100**

[0062] Effects of the work management system **100** according to one embodiment will be described below. In the work management system **100**, as will be described in a first effect to a fourth effect below, it is possible to connect pieces of information on apparatuses (for example, data collaboration, centralized data management, data integration, or translation of data) without physically modifying the apparatuses to connect each of the apparatuses to a network.

1-3-4-1. First Effect

[0063] In the work management system **100**, the server device **10** is able to notify the worker W of the work instruction that is generated based on the SOP even in an environment in which the controller **30** is not connected to a network. Furthermore, in the work management system **100**, the server device **10** is able to give a notice of the error message when the worker W has performed incorrect operation.

1-3-4-2. Second Effect

[0064] In the work management system **100**, even in an environment in which the controller **30** is not connected to a network, it is possible to digitalize data that is used for work record, record of measurement data, or the like, so that it is possible to expect data utilization.

1-3-4-3. Third Effect

[0065] In the work management system **100**, even in an environment in which the controller **30** is not connected to a network, it is not needed to perform a system modification for connecting each of the controllers **30** to a network, so that a modification period, a modification cost, a burden on the work site, or the like is not needed.

1-3-4-4. Fourth Effect

[0066] In the work management system **100**, even when a production plant or the like includes the controller **30** that is black-boxed, it is not needed to connect to a network, so that it is possible to ensure reliability in terms of confidentiality.

2. CONFIGURATION AND PROCESS OF EACH OF APPARATUSES IN WORK MANAGEMENT SYSTEM **100**

[0067] A configuration and a process of each of the apparatuses that are included in the work management system **100** illustrated in FIG. **1** will be described below with reference to FIG. **2** to FIG. **8**. In the following, an entire configuration example and a process example of the work management system **100** according to one embodiment will be first described, and thereafter, a configuration example and a process example of the server device **10**, a configuration example and a process example of the worker terminal **20**, and a configuration example and a process example of the controller **30** will be described in detail.

2-1. Entire Configuration Example of Work Management System **100**

[0068] An entire configuration example of the work management system **100** illustrated in FIG. **1** will be described with reference to FIG. **2**. FIG. **2** is a block diagram illustrating a configuration example of each of the apparatuses that are included in the work management system **100** according to one embodiment. As illustrated in FIG. **2**, the work management system **100** includes the server device **10**, the worker terminal **20**, and the controller **30**. Furthermore, the server device **10** and the worker terminal **20** are communicably connected to each other by a communication network N, such as the Internet or a dedicated line.

[0069] The server device **10** is installed in a cloud environment, an on-premise environment, an edge environment, or the like. Further, the worker terminal **20** is carried by the worker W. Furthermore, the controller **30** is installed in a work site of a production plant or the like.

2-2. Configuration Example and Process Example of Server Device **10**

[0070] A configuration example and a process example of the server device **10** will be described with reference to FIG. **2**. The server device **10** is an information providing apparatus and includes an input unit **11**, an output unit **12**, a communication unit **13**, a storage unit **14**, and a control unit **15**.

2-2-1. Input Unit **11**

[0071] The input unit **11** controls input of various kinds of information to the server device **10**. For example, the input unit **11** is implemented by a mouse, a keyboard, or the like, and receives input of various kinds of information to the server device **10**.

2-2-2. Output Unit **12**

[0072] The output unit **12** controls output of various kinds of information from the server device

10. For example, the output unit **12** is implemented by a display or the like, and displays various kinds of information that are stored in the server device **10**.

2-2-3. Communication Unit **13**

[0073] The communication unit **13** controls data communication with a different apparatus. For example, the communication unit **13** performs data communication with each of the communication apparatuses via a router or the like. Further, the communication unit **13** is able to perform data communication with a terminal or the like (not illustrated).

2-2-4. Storage Unit **14**

[0074] The storage unit **14** stores therein various kinds of information that are referred to when the control unit **15** operates, or various kinds of information that are acquired when the control unit **15** operates. The storage unit **14** stores therein an instruction data storage unit **14***a*, an operation data storage unit **14***b*, an analysis result storage unit **14***c*, and a work report storage unit **14***d*. Here, the storage unit **14** may be implemented by, for example, a semiconductor memory device, such as a Random Access Memory (RAM) or a flash memory, or a storage device, such as a hard disk or an optical disk. Meanwhile, in the example illustrated in FIG. **2**, the storage unit **14** is arranged inside the server device **10**, but may be arranged outside the server device **10**, or it may be possible to arrange a plurality of storage units.

2-2-4-1. Instruction Data Storage Unit **14***a*

[0075] The instruction data storage unit **14***a* stores therein instruction data. For example, the instruction data storage unit **14***a* stores therein instruction data that includes a work instruction that is transmitted by a notification unit **15***a* of the control unit **15** (to be described later). An example of the data that is stored in the instruction data storage unit **14***a* will be described below with reference to FIG. **3**. FIG. **3** is a diagram illustrating an example of the instruction data storage unit **14***a* of the server device **10** according to one embodiment. In the example illustrated in FIG. **3**, the instruction data storage unit **14***a* includes items of an “operation device”, a “work process”, and a “work instruction”.

[0076] The “operation device” indicates identification information for identifying an operation device that is operated by the worker W and that is not connected to a network, and is, for example, an identification number or an identification symbol of the controller **30**. Here, the “operation device” may be a robot that includes the operation panel P or a moving object, such as a vehicle or a bicycle, rather than the controller **30**, and is not specifically limited. The “work process” indicates each of steps of a work that includes single operation or a series of operation that is performed by the worker W in relation to the operation device, and is, for example, each of steps of a work that is performed by the worker W in relation to the controller **30**. The “work instruction” is an instruction for the worker W on a work that includes single operation or a series of operation that is performed by the worker W in relation to the operation device, and is, for example, a “manual work instruction”, such as raw material input with respect to a tank that is controlled by the controller **30**, a “screen operation instruction”, such as operation on the operation panel P of the controller **30**, “automatic control display”, such as data monitoring during automatic control on the controller **30**, a “data recording instruction”, such as record of measurement data of the operation panel P of the controller **30**, a “control start instruction”, such as start of operation of the controller **30**, a “control interruption instruction, such as interruption of operation of the controller **30**, or a “control termination instruction”, such as termination of operation of the controller **30**.

[0077] Specifically, FIG. **3** illustrates an example in which, with respect to the controller **30** that is identified by a “controller #1”, data, such as {work process: “work process #1”, work instruction: “work instruction #1”}, {work process: “work process #2”, work instruction: “work instruction #2”}, and {work process: “work process #3”, work instruction: “work instruction #3”}, is stored in the instruction data storage unit **14***a*.

[0078] Meanwhile, the instruction data storage unit **14***a* may store therein, operation permission (access right) on each of the controllers **30** or a work instruction.

2-2-4-2. Operation Data Storage Unit **14***b*

[0079] The operation data storage unit **14***b* stores therein operation data. For example, the operation data storage unit **14***b* stores therein operation data that is acquired by an acquisition unit **15***b* of the control unit **15** (to be described later) and that is analyzed by an analysis unit **15***c*. An example of the data that is stored in the operation data storage unit **14***b* will be described below with reference to FIG. **4**. FIG. **4** is a diagram illustrating an example of the operation data storage unit **14***b* of the server device **10** according to one embodiment. In the example illustrated in FIG. **4**, the operation data storage unit **14***b* includes items of an "operation device", a "work process", "image data", "eye-tracking data", and "location data".

[0080] The "operation device" indicates identification information for identifying an operation device that is operated by the worker W and that is not connected to a network, and is, for example, an identification number or an identification symbol of the controller **30**. The "work process" indicates each of steps of a work that includes single operation or a series of operation that is performed by the worker W in relation to the operation device, and is, for example, each of steps of a work that is performed by the worker W in relation to the controller **30**. The "image data" is data for identifying a state of an operation screen of the operation device, and is, for example, still image data or moving image data of the operation panel P that is captured by the camera C that is arranged in the controller **30**. The "eye-tracking data" is data for identifying a line of sight of the worker W with respect to the operation screen of the operation device, and is, for example, coordinate data that indicates a region on the operation panel P that is detected by the eye-tracking sensor that is arranged in the controller **30**. The "location data" is data for identifying a location of the worker W who operates the operation device, and is, for example, coordinate data that indicates the location of the worker W in a production plant that is detected by a Global Positioning System (GPS) serving as a location sensor that is incorporated in the worker terminal **20**.

[0081] Specifically, FIG. **4** illustrates an example in which, with respect to the controller **30** that is identified by the "controller #1", data, such as {work process: "work process #1", image data: "image data #1", eye-tracking data: "eye-tracking data #1"}, {work process: "work process #2", image data: "image data #2", eye-tracking data: "eye-tracking data #2"}, and {work process: "work process #3", image data: "image data #3", eye-tracking data: "eye-tracking data #3"}, is stored in the operation data storage unit **14***b*. Further, an example is illustrated in which, with respect to the worker W who operates the controller **30** that is identified by the "controller #1", data, such as {work process: "work process #1", location data: "location data #1"}, {work process: "work process #2", location data: "location data #2"}, and {work process: "work process #3", location data: "location data #3"}, is stored in the operation data storage unit **14***b*.

2-2-4-3. Analysis Result Storage Unit **14***c*

[0082] The analysis result storage unit **14***c* stores therein an analysis result. For example, the analysis result storage unit **14***c* stores therein an analysis result that is output by the analysis unit **15***c* of the control unit **15** (to be described later). An example of the data that is stored in the analysis result storage unit **14***c* will be described below with reference to FIG. **5**. FIG. **5** is a diagram illustrating an example of the analysis result storage unit **14***c* of the server device **10** according to one embodiment. In the example illustrated in FIG. **5**, the analysis result storage unit **14***c* includes items of an "operation device", a "work process", "analysis data", and "notification data".

[0083] The "operation device" indicates identification information for identifying an operation device that is operated by the worker W and that is not connected to a network, and is, for example, an identification number or an identification symbol of the controller **30**. The "work process" indicates each of steps of a work that includes single operation or a series of operation that is performed by the worker W in relation to the operation device, and is, for example, each of steps of a work that is performed by the worker W in relation to the controller **30**. The "analysis data" indicates operation on the operation device that is performed by the worker W and that is identified from the operation data or a result of the operation, and is, for example, input operation that is performed on the operation panel P of the controller **30** by the worker W in accordance with a work instruction and appropriateness of the input operation, viewing operation that is performed on the operation panel P of the controller **30** by the worker W in accordance with a work instruction and character data in a viewed region in the operation panel P, moving operation that is performed on the operation panel P of the controller **30** by the worker W in accordance with a work instruction and appropriateness of the moving operation, or the like. The “notification data” is sentence data that indicates a notification content that is given to the worker W and that is determined by using the “analysis data”, and is, for example, a work normal message indicating that the worker W has performed correct operation, a work abnormal message (error message) indicating that the worker W has performed incorrect operation or the worker W has performed operation on the incorrect controller **30**, a work change message indicating that a work that is performed by the worker W is changed, or the like.

[0084] Specifically, FIG. **5** illustrates an example in which, with respect to the controller **30** that is identified by the “controller #1”, data, such as {work process: “work process #1”, analysis data: “analysis data #1”, notification data: “notification data #1”}, {work process: “work process #2”, analysis data: “analysis data #2”, notification data: “notification data #2”}, and {work process: “work process #3”, analysis data: “analysis data #3”, notification data: “notification data #3”}, is stored in the analysis result storage unit **14***c*.

2-2-4-4. Work Report Storage Unit **14***d*

[0085] The work report storage unit **14***d* stores therein a work report. For example, the work report storage unit **14***d* stores therein a work report that is generated by a generation unit **15***d* of the control unit **15** (to be described later). An example of the data that is stored in the work report storage unit **14***d* will be described below with reference to FIG. **6**. FIG. **6** is a diagram illustrating an example of the work report storage unit **14***d* of the server device **10** according to one embodiment. In the example illustrated in FIG. **6**, the work report storage unit **14***d* includes items of an “operation device”, a “date”, and a “work report”.

[0086] The “operation device” indicates identification information for identifying an operation device that is operated by the worker W and that is not connected to a network, and is, for example, an identification number or an identification symbol of the controller **30**. The “date” indicates a work period related to the operation device that is operated by the worker W, and is, for example, year, month, and day on which a series of work processes related to the controller **30** is performed. The “work report” is a report in which a work content related to the operation device of the worker W is recorded, and is, for example, data including sentence data, such as the identification number of the controller **30**, the name of the worker W, performed work, or recorded data, image data, or the like.

[0087] Specifically, FIG. **6** illustrates an example in which, with respect to the controller **30** that is identified by the “controller #1”, data, such as {date: “date #1”, work report: “work report #1”} or {date: “date #2”, work report: “work report #2”}, is stored in the work report storage unit **14***d*.

2-2-5. Control Unit **15**

[0088] The control unit **15** controls the entire server device **10**. The control unit **15** includes the notification unit **15***a*, the acquisition unit **15***b*, the analysis unit **15***c*, and the generation unit **15***d*. Here, the control unit **15** may be implemented by, for example, an electronic circuit, such as a Central Processing Unit (CPU) or a Micro Processing Unit (MPU), or an integrated circuit, such as an Application Specific Integrated Circuit (ASIC) or a Field Programmable Gate Array (FPGA).

2-2-5-1. Notification Unit **15***a*

[0089] The notification unit **15***a* gives a notice of various kinds of information. Meanwhile, the notification unit **15***a* may refer to various kinds of information that are stored in the storage unit **14**.

[0090] The notification unit **15***a* notifies the worker W of a work instruction on a predetermined work process related to the operation device that is operated by the worker W. Further, the notification unit **15***a* notifies the worker W of a notification content that is determined by the analysis unit **15***c*. Furthermore, the notification unit **15***a* gives a notice of the work instruction via the worker terminal **20** that is carried by the worker W. Moreover, the notification unit **15***a* notifies the worker W of, as the work instruction, operation for performing each of the work processes that are generated based on the SOP. For example, the notification unit **15***a* transmits instruction data that includes a first work instruction among pieces of instruction data including work instructions on the respective work processes that are generated by the operator based on the SOP to the worker terminal **20** that is a tablet terminal carried by the worker W, and displays the instruction data on the monitor M of the worker terminal **20**.

[0091] As a specific example, the notification unit **15***a* identifies the "work process #1" that is a first work process for the worker W who operates the controller **30** that is identified by the "controller #1", refers to, from the instruction data storage unit **14***a*, {work process: "work process #1", work instruction: "work instruction #1"} as instruction data that includes the "work instruction #1" that is the first work instruction to be given to the worker W, transmits the instruction data to the worker terminal **20** that is identified as a transmission destination, and displays the "work instruction #1" together with a flowchart that indicates a work instruction on each of the work processes on the monitor M of the worker terminal **20**.

[0092] Furthermore, the notification unit **15***a* refers to, from the analysis result storage unit **14***c*, the "analysis data #1" as analysis data that is an analysis result with respect to the "work instruction #1", identifies the "work process #2" that is a second work process for the worker W who operates the controller **30** that is identified by the "controller #1" when it is determined that the worker W has performed correct operation, refers to, from the instruction data storage unit **14***a*, {work process: "work process #2", work instruction: "work instruction #2"} as instruction data that includes the "work instruction #2" that is the second work instruction to be given to the worker W, transmits the instruction data to the worker terminal **20** that is identified as the transmission destination, and displays the "work instruction #2" together with a flowchart that indicates a work instruction on each of the work processes on the monitor M of the worker terminal **20**. In this case, the "notification data #1" as notification data that is an analysis result with respect to the "work instruction #1" is referred to from the analysis result storage unit **14***c*, the notification data is transmitted to the worker terminal **20**, and a work normal message, such as "controller is normally activated by first work" or "data read is normally completed by first work", which indicates that the worker W has performed correct operation is displayed on the monitor M of the worker terminal **20**.

[0093] Moreover, the notification unit **15***a* refers to, from the analysis result storage unit **14***c*, the "analysis data #1" as analysis data that is an analysis result with respect to the "work instruction #1", refers to, from the analysis result storage unit **14***c*, the "notification data #1" as notification data that is an analysis result with respect to the "work instruction #1" when it is determined that the worker W has performed incorrect operation, transmits the notification data to the worker terminal **20**, and displays a work abnormal message, such as "controller is not normally activated by first work" or "data read is not normally completed by first work", which indicates that the worker W has performed incorrect operation on the monitor M of the worker terminal **20**. In this case, the notification unit **15***a* may cause the worker terminal **20** to generate a warning sound or the like in addition to causing the worker terminal **20** to display the work abnormal message.

[0094] Furthermore, the notification unit **15***a* refers to, from the analysis result storage unit **14***c*, the "analysis data #1" as analysis data that is an analysis result with respect to the "work instruction #1", refers to, from the instruction data storage unit **14***a*, {work process: "work process #1", work instruction: "work instruction #1-2"} as instruction data that includes a "work instruction #1-2" that is changed from the first work instruction and that is to be given to the worker W when it is determined that the worker W has performed incorrect operation, transmits the instruction data to the worker terminal **20** that is identified as the transmission destination, and displays the "work instruction #1-2” together with a flowchart that indicates a work instruction on each of the work processes on the monitor M of the worker terminal **20**. In this case, the notification unit **15***a* refers to, from the analysis result storage unit **14***c*, the “notification data #1” as notification data that is an analysis result with respect to the “work instruction #1”, transmits the notification data to the worker terminal **20**, and displays a work change message, such as “controller is not normally activated by first work, so press operation button again” or “data read is not normally completed by first work, so directly input data in tablet terminal”, which indicates that a work to be performed by the worker W is changed on the monitor M of the worker terminal **20**.

[0095] Meanwhile, as will be described in a first modification of one embodiment below, the notification unit **15***a* is able to give a notice of the work instruction via a wearable device, such as smart glasses G, which is worn on the worker W. Further, as will be described in a third modification and a fourth modification of one embodiment below, the notification unit **15***a* is able to notify the worker W of a work instruction that is transmitted by a higher-level server device **40** serving as a management apparatus that manages a plurality of operation devices. Furthermore, as will be described in a first application example of one embodiment below, the notification unit **15***a* is able to notify the worker W of a history of operation that is selected as best operation among operation that is performed in the past on the operation device, such as the controller **30**.

2-2-5-2. Acquisition Unit **15***b*

[0096] The acquisition unit **15***b* acquires various kinds of information. Meanwhile, the acquisition unit **15***b* may store the acquired various kinds of information in the storage unit **14**. Further, the acquisition unit **15***b* may refer to various kinds of information that are stored in the storage unit **14**. An operation data acquisition process (an image data acquisition process, an eye-tracking data acquisition process, and a location data acquisition process) will be described below.

Operation Data Acquisition Process

[0097] The acquisition unit **15***b* acquires operation data for identifying operation that is performed on the operation device by the worker W in accordance with a work instruction. For example, as will be described later, the acquisition unit **15***b* acquires, as the operation data, image data, eye-tracking data, location data, or the like.

Image Data Acquisition Process

[0098] The acquisition unit **15***b* acquires, as the operation data, image data for identifying a state of the operation screen of the operation device. For example, the acquisition unit **15***b* acquires image data that is captured by the camera C as an image capturing device that is arranged at a position at which an image of the operation screen of the operation device can be captured. Specifically, the acquisition unit **15***b* acquires image data that is captured by the camera C that is arranged at a position at which an image of the operation panel P of the controller **30** can be captured in the vicinity of the controller **30**.

[0099] As a specific example of the image data acquisition process, the acquisition unit **15***b* acquires, as image data that is captured by the camera C that is arranged in the vicinity of the controller **30** that is identified by the “controller #1”, {work process: “work process #1”, image data: “image data #1”}, {work process: “work process #2”, image data: “image data #2”}, {work process: “work process #3”, image data: “image data #3”}, or the like, and stores the image data in the operation data storage unit **14***b*.

Eye-Tracking Data Acquisition Process

[0100] The acquisition unit **15***b* acquires, as the operation data, eye-tracking data for identifying a line of sight of the worker W with respect to the operation screen of the operation device. For example, the acquisition unit **15***b* acquires eye-tracking data that is detected by the eye-tracking sensor S as a sensor device that is arranged at a position at which the line of sight of the worker W who operates the operation device is detectable. Specifically, the acquisition unit **15***b* acquires the eye-tracking data that is detected by the eye-tracking sensor S that is arranged at a position at which the line of sight of the worker W who operates the controller **30** is detectable in the vicinity of the controller **30**.

[0101] As a specific example of the eye-tracking data acquisition process, the acquisition unit **15***b* acquires, as the eye-tracking data that is detected by the eye-tracking sensor S that is arranged in the vicinity of the controller **30** that is identified by the “controller #1”, {work process: “work process #1”, eye-tracking data: “eye-tracking data #1”}, {work process: “work process #2”, eye-tracking data: “eye-tracking data #2”}, {work process: “work process #3”, eye-tracking data: “eye-tracking data #3”}, or the like, and stores the eye-tracking data in the operation data storage unit **14***b*.

Location Data Acquisition Process

[0102] The acquisition unit **15***b* acquires, as the operation data, location data for identifying a location of the worker W who operates the operation device. For example, the acquisition unit **15***b* acquires the location data that is detected by a location sensor as a sensor apparatus that is incorporated in the worker terminal **20**.

[0103] As a specific example of the location data acquisition process, the acquisition unit **15***b* acquires, as location data that is detected by a GPS sensor that is incorporated in the worker terminal **20** that is carried by the worker W who operates the controller **30** that is identified by the “controller #1”, {work process: “work process #1”, location data: “location data #1”}, {work process: “work process #2”, location data: “location data #2”}, {work process: “work process #3”, location data: “location data #3”}, or the like, and stores the location data in the operation data storage unit **14***b*.

[0104] Meanwhile, as will be described in the first modification of one embodiment, the acquisition unit **15***b* is able to acquire image data that is captured by a wearable device, such as the smart glasses G, which is worn on the worker W. Further, similarly, the acquisition unit **15***b* is able to acquire eye-tracking data that is captured by a wearable device, such as the smart glasses G, which is worn on the worker W. Furthermore, similarly, the acquisition unit **15***b* is able to acquire location data that is captured by a wearable device, such as the smart glasses G, which is worn on the worker W. Moreover, as will be described in a second modification of one embodiment, the acquisition unit **15***b* is able to acquire image data that is captured by the worker terminal **20**, such as a smartphone, which is carried by the worker W. Furthermore, as will be described in the third modification and the fourth modification of one embodiment, the acquisition unit **15***b* is able to acquire apparatus identification information for identifying the operation device.

2-2-5-3. Analysis Unit **15***c*

[0105] The analysis unit **15***c* analyzes various kinds of information. Meanwhile, the analysis unit **15***c* may store an analysis result in the storage unit **14**. Further, the analysis unit **15***c* may refer to various kinds of information that are stored in the storage unit **14**.

[0106] The analysis unit **15***c* analyzes the operation data and determines a notification content for the worker W. For example, the analysis unit **15***c* determines, as the notification content, a work instruction on a next work process of a predetermined work process, an error message for operation that is performed by the worker W, or a change in the work instruction on a predetermined work process.

[0107] Further, the analysis unit **15***c* determines appropriateness of operation that is performed on the operation device by the worker W by using at least one of the image data and the eye-tracking data. In this case, the analysis unit **15***c* may determine the appropriateness of the operation that is performed on the operation device by the worker W by further using the location data. Furthermore, the analysis unit **15***c* records characters of an operation screen that is viewed by the worker W by using at least one of the image data and the eye-tracking data. In this case, the analysis unit **15***c* may record the characters of the operation screen that is viewed by the worker W by further using the location data.

[0108] Here, the analysis unit **15***c* is able to generate analysis data indicating a time that is taken before a numerical value that is displayed on the operation panel P is changed from a certain value to another value, for example. Further, the analysis unit **15***c* is able to perform a process of calculating an average value when measurement data varies (is unstable), for example.

[0109] As a specific example, the analysis unit **15***c* refers to, from the operation data storage unit **14***b*, {work process: "work process #1", image data: "image data #1", eye-tracking data: "eye-tracking data #1", location data: "location data #1"} as operation data that corresponds to the "work instruction #1" that is the first work instruction that is given to the worker W who operates the controller **30** that is identified by the "controller #1". Further, the analysis unit **15***c* analyzes the "image data #1" by referring to layout information on the operation panel P (for example, button arrangement on the operation panel P) or information on response or transition of a screen (for example, "operation" lamp is lighted" when an "operation" button is pressed), which is stored in the storage unit **14**, determines whether the operation of the worker W is correct or the operation of the worker W is incorrect, and identifies the operation of the worker W or a result of the operation. Furthermore, the analysis unit **15***c* generates the "analysis data #1" as analysis data that includes the identification result. Moreover, the analysis unit **15***c* stores the "analysis data #1" as the generated analysis data in the analysis result storage unit **14***c*.

[0110] Furthermore, the analysis unit **15***c* refers to, from the operation data storage unit **14***b*, {work process: "work process #1", image data: "image data #1", eye-tracking data: "eye-tracking data #1", location data: "location data #1"} as the operation data that corresponds to the "work instruction #1" that is the first work instruction that is given to the worker W who operates the controller **30** that is identified by the "controller #1". Moreover, the analysis unit **15***c* analyzes, by an OCR process, the "image data #1" that corresponds to the coordinate data that indicates a region on the operation panel P indicated by the "eye-tracking data #1", acquires character data in the region in the operation panel P that is viewed by the worker W, and identifies the operation of the worker W or a result of the operation. Furthermore, the analysis unit **15***c* generates the "analysis data #1" as analysis data that includes the identification result. Moreover, the analysis unit **15***c* stores the "analysis data #1" as the generated analysis data in the analysis result storage unit **14***c*.

[0111] Furthermore, the analysis unit **15***c* refers to, from the operation data storage unit **14***b*, {work process: "work process #1", image data: "image data #1", eye-tracking data: "eye-tracking data #1", location data: "location data #1"} as the operation data that corresponds to the "work instruction #1" that is the first work instruction that is given to the worker W who operates the controller **30** that is identified by the "controller #1". Moreover, the analysis unit **15***c* analyzes the "location data #1" by referring to coordinate data in the production plant of the controller **30**, which is stored in the storage unit **14**, determines whether the controller **30** or a work place on or in which the worker W performs operation is correct or the controller **30** or the work place on or in which the worker W performs operation is incorrect, and identifies the operation of the worker W or a result of the operation. Furthermore, the analysis unit **15***c* generates the "analysis data #1" as analysis data that includes the identification result. Moreover, the analysis unit **15***c* stores the "analysis data #1" as the generated analysis data in the analysis result storage unit **14***c*.

[0112] Furthermore, the analysis unit **15***c* refers to, from the analysis result storage unit **14***c*, the "analysis data #1" as the analysis data, and generates, as notification data, the "notification data #1" that includes a work normal message, such as "controller is normally activated by first work" or "data read is normally completed by first work", which indicates that the worker W has performed correct operation. Moreover, the analysis unit **15***c* stores the "notification data #1" as the generated notification data in the analysis result storage unit **14***c*.

[0113] Furthermore, the analysis unit **15***c* refers to, from the analysis result storage unit **14***c*, the "analysis data #1" as the analysis data, and generates, as notification data, the "notification data #1" that includes a work abnormal message, such as "controller is not normally activated by first work" or "data read is not normally completed by first work", which indicates that the worker W has performed incorrect operation. Moreover, the analysis unit **15***c* stores the "notification data #1" as the generated notification data in the analysis result storage unit **14***c*.

[0114] Moreover, the analysis unit **15**_c_ refers to, from the analysis result storage unit **14**_c_, the "analysis data #1" as the analysis data, and generates, as notification data, the "notification data #1-2" that includes a work change message, such as "controller is not normally activated by first work, so press operation button again" or "data read is not normally completed by first work, so directly input data in tablet terminal", which indicates that a work to be performed by the worker W is changed. Furthermore, the analysis unit **15**_c_ stores the "notification data #1-2" as the generated notification data in the analysis result storage unit **14**_c_.
[0115] Meanwhile, as will be described in the third modification and the fourth modification of one embodiment below, the analysis unit **15**_c_ generates a model that enables data translation related to the operation device that is identified by using the acquired apparatus identification information. Here, the data translation is, for example, a process for enabling smooth data communication by a certain process, such as data analysis, data exchange, or data integration, between a higher-level device and a lower-level apparatus.
2-2-5-4. Generation Unit **15**_d_
[0116] The generation unit **15**_d_ generates various kinds of information. Meanwhile, the generation unit **15**_d_ may store the generated various kinds of information in the storage unit **14**. Further, the generation unit **15**_d_ may refer to various kinds of information that are stored in the storage unit **14**.
[0117] When all of the work processes performed by the worker W are completed, the generation unit **15**_d_ generates a work report in which a work status of the worker W related to the operation device is recorded. For example, when all of the work processes that are performed by the controller **30** are completed, the generation unit **15**_d_ generates a work report that includes the identification number of the controller **30**, the name of the worker W, performed work, recorded data, or the like.
[0118] As a specific example, the generation unit **15**_d_ refers to, from the analysis result storage unit **14**_c_, "analysis data #N" as analysis data that corresponds to a "work instruction #N" that is a last work instruction given to the worker W who operates the controller **30** that is identified by the "controller #1". Further, when an identification result indicating that the "work instruction #N" is normally completed is included, the generation unit **15**_d_ refers to, from the analysis result storage unit **14**_c_, the "analysis data #1", the "analysis data #2", the "analysis data #3", . . . , the "analysis data #N" as pieces of analysis data corresponding to all of the work processes, and extracts, as the identification result, the operation of the worker W or a result of the operation as a work content (the identification number of the controller **30**, the name of the worker W, performed work, recorded data, or the like). Furthermore, the generation unit **15**_d_ stores {date: "date #1", work report: "work report #1"} as a work report that includes the extracted work content in the work report storage unit **14**_d_.
2-3. Configuration Example and Process Example of Worker Terminal **20**
[0119] Referring back to FIG. **2**, a configuration example and a process example of the worker terminal **20** will be described. The worker terminal **20** is an example of an apparatus that is operated by the worker W and that is connected to a network. The worker terminal **20** includes a control unit **21**, a communication unit **22**, and a monitor M.
2-3-1. Control Unit **21**
[0120] The control unit **21** transmits various kinds of information. For example, the control unit **21** transmits the location data that is detected by the location sensor to the server device **10**. Further, the control unit **21** transmits a work instruction execution completion request that is received from the worker W, a work instruction skip request, and input data, such as measurement data, to the server device **10**.
[0121] The control unit **21** receives various kinds of information. For example, the control unit **21** receives instruction data that is transmitted from the server device **10**. Furthermore, the control unit **21** receives notification data that is transmitted from the server device **10**.
2-3-2. Communication Unit **22**

[0122] The communication unit **22** controls data communication with a different apparatus. For example, the communication unit **22** performs data communication with each of the communication apparatuses via a router or the like. Further, the communication unit **22** is able to perform data communication with a terminal or the like (not illustrated).

2-3-3. Monitor M

[0123] The monitor M controls input of various kinds of information to the worker terminal **20**. For example, the monitor M is implemented by a touch panel or the like, and receives input of various kinds of information to the worker terminal **20**. Further, the monitor M controls display of various kinds of information from the worker terminal **20**. For example, the monitor M is implemented by a touch panel or the like, and displays various kinds of information that are stored in the worker terminal **20**.

[0124] Furthermore, the monitor M displays, as a work instruction screen, the instruction data that is transmitted from the server device **10** that is the information providing apparatus. Meanwhile, details of the work instruction screen will be described below in (2-3-4. Specific example of display screen of worker terminal **20**).

2-3-4. Specific Example of Display Screen of Worker Terminal **20**

[0125] The work instruction screen will be described below with reference to FIG. **7** as a specific example of a display screen that is output by the monitor M of the worker terminal **20**. FIG. **7** is a diagram illustrating a specific example of the display screen of the worker terminal **20** according to one embodiment. A display mode of the work instruction, a type of the work instruction, and the like will be described below.

2-3-4-1. Display Mode of Work Instruction

[0126] The worker terminal **20** displays, in the work instruction screen, a flowchart that indicates a work instruction on each of the work processes on the monitor M. In the example illustrated in FIG. **7**, the worker terminal **20** notifies the worker W of the work instructions, that is, a “manual work 1 instruction”, a “screen operation 1 instruction”, and an “automatic control 1 display” in this order. Furthermore, the worker terminal **20** is able to display a work instruction that is being executed in the flowchart in a colored manner or in an enlarged manner, or display a work instruction that is already executed or a work instruction that is not yet executed in a grayed-out manner or in a reduced manner. In the example illustrated in FIG. **7**, the worker terminal **20** displays, in a grayed-out manner, the “manual work 1 instruction” that is already executed and the “automatic control 1 display” that is not yet executed, and displays, in an enlarged manner, the “screen operation 1 instruction” that is being executed. Moreover, the worker terminal **20** may display a list of the work instructions of all of the work processes, and displays, in a highlighted manner, the work instruction that is being executed.

2-3-4-2. Type of Work Instruction

[0127] The worker terminal **20** displays, in the work instruction screen, a “manual work instruction”, such as raw material input in a production plant, a “screen operation instruction”, such as operation on the operation panel P of the controller **30**, “automatic control display”, such as data monitoring during automatic control on the controller **30**, a “data recording instruction”, such as record of measurement data of the operation panel P of the controller **30**, a “control start instruction”, such as start of operation of the controller **30**, a “control interruption instruction, such as interruption of operation of the controller **30**, or a “control termination instruction”, such as termination of operation of the controller **30**, but a type of the work instruction is not specifically limited.

2-3-4-3. Others

[0128] The worker terminal **20** is able to display, in the work instruction screen, a flowchart that indicates a work instruction of each of the work processes on the monitor M, and also output, by voice, a work instruction that is being executed. Further, when the worker W taps, in the work instruction screen, the work instruction that is indicated by the flowchart on the monitor M, the worker terminal **20** is able to display a display screen that represents details of the work instruction. Furthermore, the worker terminal **20** is able to receive, in the work instruction screen, input of work instruction execution completion or input of work instruction skip by operation that is performed on the monitor M by the worker W.

2-4. Configuration Example and Process Example of Controller **30**

[0129] Referring back to FIG. **2**, a configuration example and a process example of the controller **30** will be described. The controller **30** is an example of an operation device that is operated by the worker W and that is not connected to a network. The controller **30** is an apparatus that controls a plurality of control target apparatuses and includes a control unit **31** and the operation panel P.

2-4-1. Control Unit **31**

[0130] The control unit **31** performs various kinds of control. For example, the control unit **31** transmits a control signal to a control target apparatus. Further, the control unit **31** collects measurement data from the control target apparatus.

2-4-2. Operation Panel P

[0131] The operation panel P controls input of various kinds of information to the controller **30**. For example, the operation panel P is implemented by a touch panel or the like, and receives input of tap operation that is performed on the controller **30** by the worker W. Further, the operation panel P controls display of various kinds of information from the controller **30**. For example, the operation panel P is implemented by a touch panel or the like, and displays an operation screen that includes an operation button or measurement data that is stored in the controller **30**. Meanwhile, details of the operation screen of the operation panel P will be described below in (2-4-3. Specific example of display screen of controller **30**).

2-4-3. Specific Example of Display Screen of Controller **30**

[0132] The operation screen will be described below with reference to FIG. **8** as a specific example of a display screen that is output by the operation panel P of the controller **30**. FIG. **8** is a diagram illustrating a specific example of the display screen of the controller **30** according to one embodiment. Basic display, data display, state display, and the like of the operation screen will be described below.

2-4-3-1. Basic Display of Operation Screen

[0133] As illustrated in FIG. **8**(**1**), the controller **30** displays, on the operation panel P, an operation screen that includes icons indicating an operation button, state display, data display, and the like. In the example illustrated in FIG. **8**(**1**), the controller **30** displays, as operation buttons, an icon of "operation" for receiving input of start of operation of the controller **30**, and an icon of "stop" for receiving input of termination of the operation or interruption of the operation of the controller **30**. Further, the controller **30** displays, as the state display, an icon of an "operation lamp" that is lighted when the controller **30** is operating. Furthermore, the controller **30** displays, as the data display, an icon of "data" including a numerical value of measurement data that is measured by the controller **30**. Here, the worker W is able to start operation of the controller **30** by performing tap operation on the operation button of "operation".

2-4-3-2. Data Display

[0134] As illustrated in FIG. **8**(**2**), the controller **30** displays, on the operation panel P, an operation screen that includes an icon of data display or the like. Here, when the worker W views the numerical value of the measurement data that is included in the icon of "data" for a certain period of time, the server device **10** is able to identify that the measurement data is "0000" (see a dashed ellipse in FIG. **8**(**2**)) based on the image data that is generated by the camera C or the eye-tracking data that is generated by the eye-tracking sensor S.

2-4-3-3. State Display

[0135] As illustrated in FIG. **8**(**3**), the controller **30** displays, on the operation panel P, an operation screen that includes an icon of state display or the like. Here, when the worker W performs tap operation on the operation button of "operation", the server device **10** is able to identify lighting of the “operation lamp” (see a dashed circle in FIG. **8**(**3**)) based on the image data that is generated by the camera C, and determine that the worker W has performed appropriate operation.

2-5. Camera C

[0136] Referring back to FIG. **2**, a configuration example and a process example of the camera C will be described. The camera C is an image capturing device that is installed at a position at which an image of the operation screen of the operation device can be captured. For example, the camera C is arranged at a position at which an image of the operation panel P of the controller **30** can be captured in the vicinity of the controller **30**. Further, the camera C captures an image of the operation screen of the operation panel P, and generates image data for identifying a state of the operation screen of the operation panel P.

[0137] Meanwhile, as will be described in the first modification of one embodiment, the camera C may be arranged on the smart glasses G. Further, as will be described in the second modification of one embodiment, the camera C may be mounted on the worker terminal **20**.

2-6. Eye-Tracking Sensor S

[0138] Referring back to FIG. **2**, a configuration example and a process example of the eye-tracking sensor S will be described. The eye-tracking sensor S is a sensor device that is arranged at a position at which a line of sight of the worker W who operates the operation device is detectable. For example, the eye-tracking sensor S is arranged at a position at which a line of sight of the worker W who operates the controller **30** in the vicinity of the controller **30**. Further, the eye-tracking sensor S detects a line of sight of the worker W who operates the operation panel P, and generates eye-tracking data for identifying the line of sight of the worker W with respect to the operation screen of the operation panel P.

[0139] Meanwhile, as will be described in the first modification of one embodiment, the eye-tracking sensor S may be mounted on the smart glasses G.

3. FLOW OF EACH OF PROCESSES OF WORK MANAGEMENT SYSTEM **100**

[0140] Flows of processes that are performed by the work management system **100** according to one embodiment will be described below with reference to FIG. **9** to FIG. **13**. In the following, the flow of the entire process performed by the work management system **100** will be first described, and thereafter, a work instruction notification process, an operation data acquisition process, an operation data analysis process, and a work report generation process will be described as each of processes.

3-1. Entire Process of Work Management System **100**

[0141] The flow of an entire process performed by the work management system **100** according to one embodiment will be described below with reference to FIG. **9**. FIG. **9** is a flowchart illustrating an example of the entire flow of the work management system **100** according to one embodiment. Meanwhile, processes from Step S**101** to Step S**105** below may be performed in a different order. Further, some of the processes from Step S**101** to Step S**105** below may be omitted.

3-1-1. Operation Data Acquisition Process

[0142] Firstly, the server device **10** performs the operation data acquisition process (Step S**101**). For example, by performing processes from Step S**201** to Step S**205** to be described later, the server device **10** acquires operation data for identifying operation related to the controller **30**.

3-1-2. Operation Data Analysis Process

[0143] Secondly, the server device **10** performs the operation data analysis process (Step S**102**). For example, by performing processes from Step S**301** to Step S**306** to be described later, the server device **10** identifies operation that is performed on the controller **30** by the worker W or a result of the operation, and determines a notification content for the worker W.

3-1-3. Work Instruction Notification Process

[0144] Thirdly, the server device **10** performs the work instruction notification process (Step S**103**). For example, by performing processes from Step S**401** to Step S**404** to be described later, the server device **10** notifies the worker W who operates the controller **30** of a work instruction on a

predetermined work process.
[0145] Here, when the server device **10** completes notification of the work instructions on all of the work processes (Step S**104**: Yes), the process goes to Step S**105**. In contrast, when notification of the work instructions on all of the work processes is not completed (Step S**104**: No), the process returns to Step S**101**.
3-1-4. Work Report Generation Process
[0146] Fourthly, the server device **10** performs a work report generation process (Step S**105**), and the entire process of the work management system **100** is terminated. For example, by performing processes from Step S**501** to Step S**503** to be described later, the server device **10** generates a work report in which a work content for the worker W related to the controller **30** is recorded.
3-2. Operation Data Acquisition Process
[0147] The flow of the operation data acquisition process performed by the work management system **100** according to one embodiment will be described below with reference to FIG. **10**. FIG. **10** is a flowchart illustrating an example of the flow of the operation data acquisition process of the work management system **100** according to one embodiment. Meanwhile, processes from Step S**201** to Step S**205** below may be performed in a different order. Further, some of the processes from Step S**201** to Step S**205** below may be omitted.
3-2-1. Device Operation Process
[0148] Firstly, the worker W performs the device operation process (Step S**201**). For example, the worker W performs, as operation related to the controller **30**, raw material input to a tank that is controlled by the controller **30**, input to the operation panel P of the controller **30**, visual check of measurement data that is measured by the controller **30**, or the like.
3-2-2. Image Data Acquisition Process
[0149] Secondly, the server device **10** performs the image data acquisition process (Step S**202**). For example, the server device **10** acquires the image data of the operation screen of the operation panel P that is captured by the camera C that is arranged at a position at which an image of the operation panel P of the controller **30** can be captured.
3-2-3. Eye-Tracking Data Acquisition Process
[0150] Thirdly, the server device **10** performs the eye-tracking data acquisition process (Step S**203**). For example, the server device **10** acquires eye-tracking data of the worker W with respect to the operation screen of the operation panel P, which is detected by the eye-tracking sensor S that is arranged at a position at which the line of sight of the worker W who operates the controller **30** is detectable on the operation panel P.
3-2-4. Location Data Acquisition Process
[0151] Fourthly, the server device **10** performs the location data acquisition process (Step S**204**). For example, the server device **10** acquires location data of the worker W that is detected by the location sensor that is incorporated in the worker terminal **20**.
3-2-5. Operation Data Storage Process
[0152] Fifthly, the server device **10** performs the operation data storage process (Step S**205**), and terminates the operation data acquisition process. For example, the server device **10** stores the operation data that is acquired through the processes from Step S**202** to Step S**204** in the operation data storage unit **14***b*.
3-3. Operation Data Analysis Process
[0153] The flow of the operation data analysis process performed by the work management system **100** according to one embodiment will be described below with reference to FIG. **11**. FIG. **11** is a flowchart illustrating an example of the flow of the operation data analysis process of the work management system **100** according to one embodiment. Meanwhile, processes from Step S**301** to Step S**306** below may be performed in a different order. Further, some of the processes from Step S**301** to Step S**306** below may be omitted.
3-3-1. Operation Data Reference Process

[0154] Firstly, the server device **10** performs an operation data reference process (Step S**301**). The server device **10** refers to latest operation data among pieces of operation data that are stored in the operation data storage unit **14***b*.

3-3-2. Apparatus Operation Identification Process

[0155] Secondly, the server device **10** performs an apparatus operation identification process (Step S**302**). For example, the server device **10** analyzes the operation data that is referred to in the process at Step S**301**, and identifies operation that is performed by the worker W or a result of the operation.

3-3-3. Analysis Data Generation Process

[0156] Thirdly, the server device **10** performs an analysis data generation process (Step S**303**). For example, the server device **10** generates analysis data that includes the operation that is performed by the worker W or the result of the operation, which is identified in the process at Step S**302**.

3-3-4. Notification Data Generation Process

[0157] Fourthly, the server device **10** performs a notification data generation process (Step S**304**). For example, the server device **10** determines a notification content for the worker W based on the analysis data that is generated in the process at Step S**303**, and generates notification data that includes a message indicating the notification content.

3-3-5. Notification Data Transmission Process

[0158] Fifthly, the server device **10** performs a notification data transmission process (Step S**305**). For example, the server device **10** transmits the notification data that is generated in the process at Step S**304** to the worker terminal **20**.

3-3-6. Analysis Result Storage Process

[0159] Sixthly, the server device **10** performs an analysis result storage process (Step S**306**), and terminates the operation data analysis process. For example, the server device **10** stores the analysis data that is generated in the process at Step S**303** and the notification data that is generated in the process at Step S**304** in the analysis result storage unit **14***c*.

3-4. Work Instruction Notification Process

[0160] The flow of the work instruction notification process performed by the work management system **100** according to one embodiment will be described below with reference to FIG. **12**. FIG. **12** is a flowchart illustrating an example of the flow of the work instruction notification process of the work management system **100** according to one embodiment. Meanwhile, processes from Step S**401** to Step S**404** below may be performed in a different order. Further, some of the processes from Step S**401** to Step S**404** may be omitted.

3-4-1. Work Process Identification Process

[0161] Firstly, the server device **10** performs a work process identification process (Step S**401**). For example, the server device **10** identifies a work process for which a work instruction is to be given to the worker W among all of the work processes for performing works on the controller **30**.

3-4-2. Instruction Data Reference Process

[0162] Secondly, the server device **10** performs an instruction data reference process (Step S**402**). The server device **10** refers to instruction data that includes the work instruction on the work process that is identified in the process at Step S**401** among the work instructions that are stored in the instruction data storage unit **14***a*.

3-4-3. Instruction Data Transmission Process

[0163] Thirdly, the server device **10** performs the instruction data transmission process (Step S**403**). For example, the server device **10** transmits instruction data that includes the work instruction that is referred to in the process at Step S**402** to the worker terminal **20**.

3-4-4. Work Instruction Display Process

[0164] Fourthly, the worker terminal **20** performs a work instruction display process (Step S**404**), and terminates the work instruction notification process. For example, the worker terminal **20** receives the instruction data that includes the work instruction that is transmitted in the process at

Step S**403**, displays the work instruction to be given to the worker W in a highlighted manner, and displays a flowchart or a list that indicates the work instruction on each of the work processes on the monitor M.

3-5. Work Report Generation Process

[0165] The flow of the work report generation process performed by the work management system **100** according to one embodiment will be described below with reference to FIG. **13** FIG. **13** is a flowchart illustrating an example of the operation data analysis process of the work management system **100** according to one embodiment. Meanwhile, processes from Step S**501** to Step S**503** below may be performed in a different order. Further, some of the processes from Step S**501** to Step S**503** may be omitted.

3-5-1. Analysis Result Reference Process

[0166] Firstly, the server device **10** performs an analysis result reference process (Step S**501**). For example, when all of the work processes performed by the worker W are terminated, the server device **10** refers to analysis data corresponding to a target period among pieces of analysis data that are stored in the analysis result storage unit **14***c*.

3-5-2. Work Content Extraction Process

[0167] Secondly, the server device **10** performs a work content extraction process (Step S**502**). For example, the server device **10** extracts a work content from the analysis data that is referred to in the process at Step S**501**, and outputs a work report that includes the extracted work content.

3-5-3. Work Report Storage Process

[0168] Thirdly, the server device **10** performs a work report storage process (Step S**503**), and terminates the work report generation process. For example, the server device **10** stores the work report that is output in the process at Step S**502** in the work report storage unit **14***d*.

4. Modifications and Application Examples of One Embodiment

[0169] Modifications and application examples of one embodiment will be described below with reference to FIG. **14** to FIG. **18**. In the following, a first modification to a fourth modification and a first application example to a seventh application example of one embodiment will be described.

4-1. First Modification of One Embodiment

[0170] A configuration and a process of a work management system **100**M-**1** according to the first modification of one embodiment will be described below with reference to FIG. **14** and FIG. **15**. FIG. **14** is a diagram illustrating a configuration example and a process example of the work management system **100**M-**1** according to the first modification of one embodiment. FIG. **15** is a diagram illustrating a specific example of a display screen of the smart glasses G according to the first modification of one embodiment. Meanwhile, explanation of the same configuration and the same processes as those of one embodiment will be omitted.

4-1-1. Configuration Example of Work Management System **100**M-**1**

[0171] The work management system **100**M-**1** includes the server device **10**, the smart glasses G, and the controller **30**. The server device **10** and the smart glasses G are communicably connected to each other in a wired or wireless manner via a predetermined communication network (not illustrated). Meanwhile, the work management system **100**M-**1** may include a relay apparatus (not illustrated) for connecting the server device **10** and the smart glasses G to a predetermined communication network.

[0172] The smart glasses G is a wearable device that is worn on the worker W in a work site of a production plant. The smart glasses G includes the camera C that is able to capture an image of the operation screen of the operation panel P. Further, the smart glasses G has a monitor function to enable display of a work instruction and an eye-tracking sensor function to enable detection of a line of sight of the worker W.

[0173] Meanwhile, the server device **10** is the same as that of the work management system **100** according to one embodiment, and therefore, explanation thereof will be omitted. Further, the controller **30** is the same as that of the work management system **100** according to one embodiment except that the camera C and the eye-tracking sensor S are not arranged in the vicinity of the controller **30**, and therefore, explanation thereof will be omitted.

4-1-2. Process Example of Work Management System **100**M-**1**

[0174] The work management system **100**M-**1** performs a process as described below. Firstly, the worker W operates the controller **30** (Step S**11**). Secondly, the server device **10** acquires operation data via the smart glasses G (Step S**12**). Thirdly, the server device **10** analyzes the operation data (Step S**13**). Fourthly, the server device **10** notifies the worker W of a work instruction via the smart glasses G (Step S**14**). Fifthly, the server device **10** generates a work report (Step S**15**).

4-1-3. Specific Example of Display Screen of Smart Glasses G

[0175] As a specific example of the display screen that is output by the smart glasses G, a work instruction screen will be described below with reference to FIG. **15**.

[0176] As illustrated in FIG. **15**(**1**), in the work instruction screen, the smart glasses G displays the flowchart that indicates the work instruction on each of the work processes. Further, the worker W is able to recognize operation that is being performed on the operation panel P of the controller **30** in a region in which the work instruction screen is not displayed.

[0177] Specifically, when the worker W is wearing the smart glasses G, the worker W is able to view a work display screen of the smart glasses G and simultaneously perform operation on the controller **30**.

[0178] As illustrated in FIG. **15**(**2**), when the worker W views, via the smart glasses G, a numerical value of the measurement data that is included in the icon of "data" for a certain period of time or more, the server device **10** is able to identify that the measurement data is "0000" (see a dashed ellipse in FIG. **15**(**2**)) based on the image data that is generated by the camera C that is arranged on the smart glasses G or the eye-tracking data that is generated by the smart glasses G.

[0179] As illustrated in FIG. **15**(**3**), when the worker W who is wearing the smart glasses G performs tap operation on an operation button of "operation", the server device **10** is able to identify lighting of the "operation lamp" (see a dashed circle in FIG. **15**(**3**)) based on the image data that is generated by the camera C that is arranged on the smart glasses G, and determine that the worker W has performed appropriate operation.

4-1-4. Effect of Work Management System **100**M-**1**

[0180] As described above, in the work management system **100**M-**1** according to the first modification of one embodiment, it is not needed to arrange the camera C and the eye-tracking sensor S on the controller **30** of the work management system **100** according to one embodiment, and it is not needed to always carry the worker terminal **20**, such as a tablet terminal. Therefore, in the work management system **100**M-**1**, it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network and without further changing an existing facility as compared to the work management system **100**.

4-2. Second Modification of One Embodiment

[0181] A configuration and a process of a work management system **100**M-**2** according to the second modification of one embodiment will be described below with reference to FIG. **16**. FIG. **16** is a diagram illustrating a configuration example and a process example of the work management system **100**M-**2** according to the second modification of one embodiment. Meanwhile, explanation of the same configuration and the same processes as those of one embodiment will be omitted.

4-2-1. Configuration Example of Work Management System **100**M-**2**

[0182] The work management system **100**M-**2** includes the server device **10**, the worker terminal **20**, and the controller **30**. The server device **10** and the worker terminal **20** are communicably connected to each other in a wired or wireless manner via a predetermined communication network (not illustrated).

[0183] The worker terminal **20** is a terminal that is carried by the worker W in a work site of a production plant. The worker terminal **20** is implemented by a smartphone or the like, and includes the monitor M that displays a work instruction. The worker terminal **20** includes the camera C that is able to capture an image of the operation screen of the operation panel P. Further, the work management system **100**M-**2** may include the eye-tracking sensor S that is able to detect a line of sight of the worker W who operates the controller **30** and that is arranged in the vicinity of the controller **30**.

[0184] As illustrated in FIG. **16**, when operating the operation panel P, the worker W stores the worker terminal **20** in a chest pocket or the like of the worker W such that a camera lens of the camera C protrudes so as to be able to capture an image of the operation screen of the operation panel P. In contrast, when viewing the work instruction screen, the worker W takes out the worker terminal **20** from the chest pocket or the like of the worker W and uses the worker terminal **20**. In this case, the worker W may receive a work instruction by voice guidance that is output by the worker terminal **20** without taking out the worker terminal **20** from the chest pocket or the like of the worker W.

[0185] Meanwhile, the server device **10** is the same as that of the work management system **100** according to one embodiment, and therefore, explanation thereof will be omitted. Further, the controller **30** is the same as that of the work management system **100** according to one embodiment except that the camera C is not arranged in the vicinity of the controller **30**, and therefore, explanation thereof will be omitted. Furthermore, the eye-tracking sensor S may be arranged in the vicinity of the controller **30**.

4-2-2. Process Example of Work Management System **100**M-**2**

[0186] The work management system **100**M-**2** performs a process as described below. Firstly, the worker W operates the controller **30** (Step S**21**). Secondly, the server device **10** acquires operation data via the worker terminal **20** (Step S**22**). Thirdly, the server device **10** analyzes the operation data (Step S**23**). Fourthly, the server device **10** notifies the worker W of a work instruction via the worker terminal **20** (Step S**24**). Fifthly, the server device **10** generates a work report (Step S**25**).

4-2-3. Effect of Work Management System **100***m*-**2**

[0187] As described above, in the work management system **100**M-**2** according to the second modification of one embodiment, it is not needed to arrange the camera C or the like in the vicinity of the controller **30** of the work management system **100** according to one embodiment, and it is needed to carry only the worker terminal **20** that includes the camera C. Therefore, in the work management system **100**M-**2**, it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network, without further changing an existing facility as compared to the work management system **100**, and without increasing the number of terminals as compared to the work management system **100**M-**1**.

4-3. Third Modification of One Embodiment

[0188] A configuration and a process of a work management system **100**M-**3** according to the third modification of one embodiment will be described below with reference to FIG. **17**. FIG. **17** is a diagram illustrating a configuration example and a process example of the work management system **100**M-**3** according to the third modification of one embodiment. Meanwhile, explanation of the same configuration and the same processes as those of one embodiment will be omitted.

4-3-1. Configuration Example of Work Management System **100**M-**3**

[0189] The work management system **100**M-**3** includes the server device **10**, a worker terminal **20**M, the controller **30**, and the higher-level server device **40**. The server device **10**, the worker terminal **20**M, and the higher-level server device **40** are communicably connected to each other in a wired or wireless manner via a predetermined communication network (not illustrated).

[0190] The worker terminal **20**M is a terminal that is carried by the worker W in a work site of a production plant. The worker terminal **20**M is implemented by a smartphone or the like, and includes the monitor M that displays a work instruction. The worker terminal **20**M acquires the apparatus identification information for identifying the controller **30**. In this case, the worker terminal **20**M acquires the identification information, such as a barcode, on the controller **30** which is displayed on the operation panel P of the controller **30** by operation of the worker W. Further, the worker terminal **20**M is able to acquire location data of the worker W via the location sensor that is incorporated in the worker terminal **20**M, and identify the controller **30** from the acquired location data. Furthermore, the worker terminal **20**M is able to acquire the apparatus identification information that is directly input by the worker W. Specifically, when operating the controller **30**, the worker terminal **20**M is able to check whether or not the controller **30** is a correct operation target before the operation.

[0191] Furthermore, the worker terminal **20**M generates a model that enables data translation related to the operation device that is identified by using the acquired apparatus identification information. For example, the worker terminal **20**M generates an information model that enables translation of a parameter that is set in the identified controller **30** by using the acquired apparatus identification information, and performs data communication via the generated information model. Here, the information model is a model that allows the higher-level server device **40** to recognize the plurality of controllers **30** that receive different expressions and instructions as similar instructions and expressions. For example, the information model has a function to perform normalization such that even when units of numerical values are different due to a difference in vendors among the controllers **30** that have similar functions, the higher-level server device **40** can recognize the units as the same units. Therefore, with use of the information model, connection setting or the like becomes easy even when the plurality of controllers **30** cooperate with one another or a plurality of server devices **50** are connected to a higher-level device as will be described later in the fourth modification of one embodiment. Meanwhile, the process of acquiring the device identification information, the process of generating the information mode, and the process of performing the data communication via the information model as described above may be performed by the server device **10**.

[0192] The higher-level server device **40** is a management apparatus that manages the server device **10**. The higher-level server device **40** is able to notify the worker W of the work instruction via the information model that is generated by the worker terminal **20**M. Specifically, the higher-level server device **40** is able to generate not only a work instruction that is generated based on the SOP, but also a work instruction that is specialized for the specific controller **30**, a work instruction that corresponds to emergency, or the like, and notify the worker W of the work instruction.

[0193] Meanwhile, the server device **10** is the same as that of the work management system **100** according to one embodiment except for a process that is performed or that is not performed by the work management system **100**M-**3**, and therefore, explanation thereof will be omitted. Further, the controller **30** is the same as that of the work management system **100** according to one embodiment or that of the work management system **100**M-**2** according to the second modification of one embodiment, and therefore, explanation thereof will be omitted.

4-3-2. Process Example of Work Management System **100**M-**3**

[0194] The work management system **100**M-**3** performs a process as described below. Firstly, the worker terminal **20**M identifies the controller **30** that is a target apparatus by using the device identification information (Step S**31**). Secondly, the worker terminal **20**M generates an information model that translates various kinds of data of the identified controller **30** (Step S**32**). Here, the information model performs conversion of, as the various kinds of data, various kinds of parameters, measurement data that is displayed on the controller **30**, or a unit system that is used for operation of the controller **30**. Specifically, the information model is able to perform data conversion on other than a unit system in which a higher-side is represented by “High” or “Low” and a lower-side is represented by “1” or “0”. Thirdly, the worker W operates the operation panel P of the controller **30** (Step S**33**). Fourthly, the worker terminal **20**M acquires the operation data (Step S**34**). Fifthly, the worker terminal **20**M analyzes the operation data (Step S**35**). Sixthly, the worker terminal **20**M transmits an analysis result to the server device **10** and the higher-level server device **40** via the information model (Step S**36**). Seventhly, the higher-level server device **40** transmits instruction data to the server device **10** (Step S**37**). Eighthly, the server device **10** notifies the worker W of the work instruction via the information model of the worker terminal **20** (Step S**38**). Ninthly, the server device **10** generates a work report (Step S**39**).

4-3-3. Effect of Work Management System **100**M-**3**

[0195] As described above, in the work management system **100**M-**3** according to the third modification of one embodiment, it is possible to output an analysis result based on the information model that enables translation of various kinds of data of the identified controller **30**. Further, in the work management system **100**M-**3**, it is possible to notify the worker W of not only the work instruction that is generated from the SOP, but also the work instruction that is generated by the higher-level server device **40**. Therefore, the work management system **100**M-**3** makes it possible to flexibly generate a work instruction at the time of emergency or the like as compared to the work management system **100**, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

4-4. Fourth Modification of One Embodiment

[0196] A configuration and a process of a work management system **100**M-**4** according to the fourth modification of one embodiment will be described below with reference to FIG. **18**. FIG. **18** is a diagram illustrating a configuration example and a process example of the work management system **100**M-**4** according to the fourth modification of one embodiment. Meanwhile, explanation of the same configuration and the same processes as those of one embodiment will be omitted.

4-4-1. Configuration of Work Management System **100**M-**4**

[0197] The work management system **100**M-**4** includes a server device **10**M, the worker terminal **20**M, the controller **30**, the higher-level server device **40**, and a cooperation server device **50**. The server device **10**M, the worker terminal **20**M, the higher-level server device **40**, and the cooperation server device **50** are communicably connected to each other in a wired or wireless manner via a predetermined communication network (not illustrated).

[0198] The work management system **100**M-**4** is able to integrally manage the plurality of controllers **30** (**30**A, **30**B, . . . , **30**X). Further, in the work management system **100**M-**4**, it is possible to use the cooperation server device **50** that provides various kinds of applications, in addition to the higher-level server device **40** that is included in the work management system **100**M-**3** as described above. Specifically, in the work management system **100**M-**4**, it is possible to use, as the cooperation server device **50**, an engineering application server device **50**A that changes a relationship between the controllers **30** or changes a system screen of the higher-level server device **40**, an apparatus maintenance application server device **50**B that is able to recognize deterioration or a timing of exchange of the controller **30** from the image data of the controller **30**, a control application server device **50**C that enables alarm monitoring, or the like. Further, in the work management system **100**M-**4**, the server device **10**M is able to generate or convert the information model of each of the controllers **30** based on the collected device identification information.

4-4-2. Effect of Work Management System **100**M-**4**

[0199] As described above, in the work management system **100**M-**4** according to the fourth modification of one embodiment, it is possible to manage cooperation of works that are performed by the plurality of workers W (W-**1**, W-**2**, . . . , W-n) and the plurality of workers W. In other words, in the work management system **100**M-**4**, it is possible to give a notice of the work instruction of the higher-level server device **40** or the cooperation server device **50**, so that, by causing each of the workers W to perform a work in accordance with the work instruction, the workers W can cooperate with each other. For example, in the work management system **100**M-**4**, by causing the plurality of work terminals **20**M (**20**M-**1**, **20**M-**2**, . . . , **20**M-n) to directly exchange data, it is possible to perform cooperation for adjusting a priority of operation or an operation timing. Therefore, the work management system **100**M-**4** is able to manage the increased number of controllers **30** as compared to the work management system **100**, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

4-5. Application Examples of One Embodiment

[0200] Application examples of one embodiment will be described below. In the following, a first application example to a seventh application example will be described. Meanwhile, the first application example to the seventh application example of one embodiment are applicable to the first modification to the fourth modification of one embodiment as described above.

4-5-1. First Application Example

[0201] As the first application example of one embodiment, it is possible to display, on the work instruction screen of the worker terminal **20** or the like, information on data (golden batch or the like) that was excellent during past production or a trend in a superimposed manner in addition to the work instruction.

4-5-2. Second Application Example

[0202] As the second application example of one embodiment, when the worker terminal **20** or the like acquires the location data of the worker W, it is possible to keep a record of movement of the worker W between the different controllers **30**; therefore, by using this data, it is possible to perform scheduling to optimize a movement route when the plurality of workers W perform works.

4-5-3. Third Application Example

[0203] As the third application example of one embodiment, when the worker terminal **20** or the like acquires the location data of the worker W, it is possible to suggest a traffic flow of the worker to give a higher priority to health, rather than time, as a purpose of optimization. For example, it is possible to suggest a traffic flow to a certain person, for whom a lack of exercise is pointed out from the health perspective, such that an appropriate physical activity can be ensured.

4-5-4. Fourth Application Example

[0204] As the fourth application example of one embodiment, the technology is applicable to, as the worker W, a robot other than a human being.

4-5-5. Fifth Application Example

[0205] As the fifth application example of one embodiment, when the worker W and the worker terminal **20** are associated with each other by an access right or the like and an image of the operation panel P is captured, and if the captured operation panel P is not responsible for the worker W who carries the worker terminal **20**, it is possible to output warning display on the worker terminal **20**.

4-5-6. Sixth Application Example

[0206] As the sixth application example of one embodiment, when the plurality of workers W concurrently perform works in a single production line and the works need to be performed in order, it is possible to display an execution timing of each of the works on the worker terminal **20**.

4-5-7. Seventh Application Example

[0207] As the seventh application example of one embodiment, it is possible to arrange the camera C on a robot or the like that moves automatically, automatically capture an image of the operation panel P of the operation screen of each of the controllers **30**, and call the worker W who is present nearby only when treatment is needed.

5. EFFECTS OF ONE EMBODIMENT

[0208] Effects of one embodiment will be described below. First effects to fourteenth effects of the processes according to one embodiment will be described below.

5-1. First Effect

[0209] Firstly, in one process according to one embodiment as described above, the server device **10** acquires operation data for identifying operation that is performed on the controller **30** by the worker W in accordance with a work instruction on a predetermined work process related to the operation device that is operated by the worker, analyzes the operation data, determines a notification content for the worker W, and notifies the worker W of the determined notification content. Therefore, in this process, it is possible to notify the worker W of the notification content in accordance with the operation that is performed on the controller **30** that is not connected to the network, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

5-2. Second Effect

[0210] Secondly, in one process according to one embodiment as described above, the server device **10** acquires, as the operation data, image data for identifying a state of the operation screen of the controller **30**, and determines appropriateness of the operation that is performed on the controller **30** by the worker W by using the acquired image data. Therefore, in this process, it is possible to determine the appropriateness of the work that is performed by the worker W by using the image data without changing the controller **30** that is not connected to the network, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

5-3. Third Effect

[0211] Thirdly, in one process according to one embodiment as described above, the server device **10** further acquires, as the operation data, eye-tracking data for identifying a line of sight of the worker W with respect to the operation screen of the controller **30**, and determines appropriateness of operation that is performed on the controller **30** by the worker W by using the acquired image data and the acquired eye-tracking data. Therefore, in this process, it is possible to determine the appropriateness of the work that is performed by the worker W by using the image data and the eye-tracking data without changing the controller **30** that is not connected to the network, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

5-4. Fourth Effect

[0212] Fourthly, in one process according to one embodiment as described above, the server device **10** acquires, as the operation data, image data for identifying a state of the operation screen of the controller **30**, and records characters in the operation screen that is viewed by the worker W by using the acquired image data. Therefore, in this process, it is possible to digitalize data of the work that is performed by the worker W by using the image data without changing the controller **30** that is not connected to the network, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

5-5. Fifth Effect

[0213] Fifthly, in one process according to one embodiment as described above, the server device **10** further acquires, as the operation data, eye-tracking data for identifying a line of sight of the worker W with respect to the operation screen of the controller **30**, and records characters in the operation screen that is viewed by the worker W by using the acquired image data and the acquired eye-tracking data. Therefore, in this process, it is possible to digitalize data of the work that is performed by the worker W by using the image data and the eye-tracking data without changing the controller **30** that is not connected to the network, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

5-6. Sixth Effect

[0214] Sixthly, in one process according to one embodiment as described above, the server device **10** notifies the worker W of, as the work instruction, operation for performing each of work processes that are generated from the SOP. Therefore, in this process, it is possible to give a notice of the work instruction to be executed by the worker W based on the SOP in a production facility or the like in which a work content, such as batch, is frequently changed, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network, which makes it possible to notify the worker of a correct work content and prevent incorrect operation.

5-7. Seventh Effect

[0215] Seventhly, in one process according to one embodiment as described above, the server device **10** determines, as the notification content, a work instruction on a next work process of the predetermined work process, an error message for operation that is performed by the worker W, or a change of a work instruction on the predetermined work process. Therefore, in this process, it is possible to give a notice of the work instruction to be executed by the worker W in each of the work processes, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network, which makes it possible to give an instruction on correct operation to the worker W in a work site.

5-8. Eighth Effect

[0216] Eighthly, in one process according to one embodiment as described above, when all of work processes performed by the worker W are completed, the server device **10** generates a work report in which a work content for the worker W related to the controller **30** is recorded. Therefore, in this process, it is possible to automatically generate the work report every time a series of works is completed, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network, which makes it possible to automatically record a content, which has been manually recorded in a note or a work report, in the server device **10** without bothering the worker W and prevent mistakes due to erroneous description.

5-9. Ninth Effect

[0217] Ninthly, in one process according to one embodiment as described above, the server device **10** gives a notice of a work instruction via the worker terminal **20** that is carried by the worker W, and acquires the image data that is captured by the camera C that is arranged at a position at which an image of the operation screen of the controller **30** is capturable and the eye-tracking data that is detected by the eye-tracking sensor S that is arranged at a position at which a line of sight of the worker W who operates the controller **30** is detectable. Therefore, in this process, it is possible to analyze the operation that is performed by the worker W based on the operation data that is acquired from the camera C and the eye-tracking sensor S that are arranged at the controller **30**, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

5-10. Tenth Effect

[0218] Tenthly, in one process according to the first modification of one embodiment as described above, the server device **10** gives a notice of the work instruction via the smart glasses G that is worn on the worker W, and acquires the image data that is captured by the smart glasses G and the eye-tracking data that is detected by the smart glasses G. Therefore, in this process, it is possible to analyze the operation that is performed by the worker W based on the operation data that is acquired from the smart glasses G, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

5-11. Eleventh Effect

[0219] Eleventhly, in one process according to the second modification of one embodiment as described above, the server device **10** gives a notice of the work instruction via the worker terminal **20** that is carried by the worker W, and acquires the image data that is captured by the worker terminal **20**. Therefore, in this process, it is possible to analyze the operation that is performed by the worker W based on the operation data that is acquired from the worker terminal **20** that is carried by the worker W, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

5-12. Twelfth Effect

[0220] Twelfthly, in one process according to the third modification of one embodiment as described above, the server device **10** acquires device identification information for identifying the controller **30**, and generates a model that enables data translation related to the controller **30** that is identified by using the device identification information. Therefore, in this process, with use of the information model between a plurality of apparatuses or servers that use different expressions, it is possible to exchange information without recognizing a counterparty, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

5-13. Thirteenth Effect

[0221] Thirteenthly, in one process according to the fourth modification of one embodiment as described above, the server device **10** notifies the worker W of a work instruction that is transmitted by the higher-level server device **40** that manage the plurality of controllers **30**. Therefore, in this process, it is possible to analyze cooperation among the plurality of workers W, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

5-14. Fourteenthly Effect

[0222] Fourteenthly, in the first application example of one embodiment as described above, the server device **10** notifies the worker W of a history of operation that is selected as best operation among operation that is performed in the past on the controller **30**. Therefore, in this process, it is possible to refer to information, such as golden batch, and change to an appropriate state based on determination made by the worker W, so that it is possible to more effectively connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

System

[0223] The processing procedures, control procedures, specific names, and information including various kinds of data and parameters illustrated in the above-described document and drawings may be arbitrarily changed unless otherwise specified.

[0224] Further, the components of the apparatuses illustrated in the drawings are functionally conceptual and do not necessarily have to be physically configured in the manner illustrated in the drawings. In other words, specific forms of distribution and integration of the apparatuses are not limited to those illustrated in the drawings. That is, all or part of the apparatuses may be functionally or physically distributed or integrated in arbitrary units depending on various loads or use conditions.

[0225] Furthermore, all or an arbitrary part of processing functions that are implemented by the apparatuses may be realized by a CPU and a program analyzed and executed by the CPU, or may be realized by hardware using wired logic.

Hardware

[0226] A hardware configuration examples of the server device **10** that is an information providing apparatus will be described below. Meanwhile, the other apparatuses may have the same hardware configurations. FIG. **19** is a diagram illustrating a hardware configuration example according to one embodiment. As illustrated in FIG. **19**, the server device **10** includes a communication device **10***a*, a Hard Disk Drive (HDD) **10***b*, a memory **10***c*, and a processor **10***d*. Meanwhile, all of the units illustrated in FIG. **19** are connected to one another via a bus or the like.

[0227] The communication device **10***a* is a network interface card or the like and performs communication with a different server. The HDD **10***b* stores therein a program or a database for implementing the functions illustrated in FIG. **2**.

[0228] The processor **10***d* reads a program that executes the same processes as those of each of the processing units illustrated in FIG. **2** from the HDD **10***b* or the like, loads the programs onto the memory **10***c*, and operates the processes for implementing each of the functions as described above with reference to FIG. **2** or other figures. For example, the process as described above implements the same functions as those of the processing units included in the server device **10**. Specifically, the processor **10***d* reads, from the HDD **10***b* or the like, programs that have the same functions as those of the notification unit **15***a*, the acquisition unit **15***b*, the analysis unit **15***c*, the generation unit **15***d*, and the like. Further, the processor **10***d* performs processes that implement the same processes as those of the notification unit **15***a*, the acquisition unit **15***b*, the analysis unit **15***c*, the generation unit **15***d*, and the like.
[0229] In this manner, by reading and executing the program, the server device **10** operates as an apparatus that implements various kinds of processing methods. Further, the server device **10** may be able to implement the same functions as those of the embodiments as described above by causing a medium reading apparatus to read the above-described program from a recording medium and executing the read program. Meanwhile, program described in this embodiment need not always be executed by the server device **10**. For example, the present invention may be applied in the same manner to a case in which a different computer or a server executes the program and a case in which the different computer and the server execute the program in a cooperative manner.
[0230] The program as described above may be distributed via a network, such as the Internet. Further, the program as described above may be recorded in a computer readable recording medium, such as a hard disk, a flexible disk (FD), a compact disk read only memory (CD-ROM), a Magneto-Optical disk (MO), a Solid State Drive (SSD), or a Digital Versatile Disc (DVD), and may be executed by being read from the recording medium by a computer. Furthermore, the program as described above may be downloaded in a storage area that is included in the worker terminal **20**.
[0231] According to the present invention, it is possible to connect pieces of information on apparatuses without physically modifying the apparatuses to connect each of the apparatuses to a network.

Claims

1. An information providing apparatus comprising: an acquisition unit that acquires operation data for identifying operation that is performed on an operation device by a worker in accordance with a work instruction on a predetermined work process related to the operation device that is operated by the worker; an analysis unit that analyzes the operation data and determines a notification content for the worker; and a notification unit that notifies the worker of the determined notification content.
2. The information providing apparatus according to claim 1, wherein the acquisition unit acquires, as the operation data, image data for identifying a state of an operation screen of the operation device, and the analysis unit determines appropriateness of operation that is performed on the operation device by the worker by using the image data.
3. The information providing apparatus according to claim 2, wherein the acquisition unit further acquires, as the operation data, eye-tracking data for identifying a line of sight of the worker with respect to the operation screen, and the analysis unit determines appropriateness of operation that is performed on the operation device by the worker by using the image data and the eye-tracking data.
4. The information providing apparatus according to claim 1, wherein the acquisition unit acquires, as the operation data, image data for identifying a state of an operation screen of the operation device, and the analysis unit records characters in the operation screen that is viewed by the worker by using the image data.
5. The information providing apparatus according to claim 4, wherein the acquisition unit further

acquires, as the operation data, eye-tracking data for identifying a line of sight of the worker with respect to the operation screen, and the analysis unit records characters in the operation screen that is viewed by the worker by using the image data and the eye-tracking data.

6. The information providing apparatus according to claim 1, wherein the notification unit notifies the worker of, as the work instruction, operation for performing each of work processes that are generated from a standard operation procedure.

7. The information providing apparatus according to claim 1, wherein the analysis unit determines, as the notification content, one of work instructions on a next work process in the predetermined work process, an error message for operation that is performed by the worker, or a change of a work instruction on the predetermined work process.

8. The information providing apparatus according to claim 1, further comprising: a generation unit that generates, when all of work processes performed by the worker are completed, a work report in which a work content for the worker related to the operation device is recorded.

9. The information providing apparatus according to claim 3, wherein the notification unit gives a notice of the work instruction via a worker terminal that is carried by the worker, and the acquisition unit acquires the image data that is captured by an image capturing device that is arranged at a position at which an image of the operation screen of the operation device is capturable and the eye-tracking data that is detected by a sensor device that is arranged at a position at which a line of sight of the worker who operates the operation device is detectable.

10. The information providing apparatus according to claim 3, wherein the notification unit gives a notice of the work instruction via a wearable device that is worn on the worker, and the acquisition unit acquires the image data that is captured by the wearable device and the eye-tracking data that is detected by the wearable device.

11. The information providing apparatus according to claim 2, wherein the notification unit gives a notice of the work instruction via a worker terminal that is carried by the worker, and the acquisition unit acquires the image data that is captured by the worker terminal.

12. The information providing apparatus according to claim 1, wherein the acquisition unit acquires device identification information for identifying the operation device, and the analysis unit generates a model that enables data translation related to the operation device that is identified by using the device identification information.

13. The information providing apparatus according to claim 1, wherein the notification unit notifies the worker of the work instruction that is transmitted by a management apparatus that manages a plurality of operation devices.

14. The information providing apparatus according to claim 1, wherein the notification unit notifies the worker of a history of operation that is selected as best operation among operation that is performed in a past on the operation device.

15. An information providing method that is implemented by a computer, the information providing method comprising: acquiring operation data for identifying operation that is performed on an operation device by a worker in accordance with a work instruction on a predetermined work process related to the operation device that is operated by the worker; analyzing the operation data; determining a notification content for the worker; and notifying the worker of the determined notification content.

16. A non-transitory computer-readable recording medium having stored therein an information providing program that causes a computer to execute a process, the process comprising: acquiring operation data for identifying operation that is performed on an operation device by a worker in accordance with a work instruction on a predetermined work process related to the operation device that is operated by the worker; analyzing the operation data; determining a notification content for the worker; and notifying the worker of the determined notification content.